

# NLP Assignment 2

**Jana Nikolovska, Marco Sangiorgi, and Andrea Fossà**

Master's Degree in Artificial Intelligence, University of Bologna  
{ jana.nikolovska, marco.sangiorgi24, andrea.fossa }@studio.unibo.it

## Abstract

This assignment focuses on the implementation of open source Large Language Models (LLMs) with prompting approaches to tackle the task of detecting sexism in tweets. In this report we will evaluate the classification performance and compare the performances of different pre-trained LLMs by using both zero- and few-shot-prompting techniques.

## 1 Introduction

Sexism online is a significant issue, contributing to harm, exclusion, and the reinforcement of societal inequalities. Automated tools can aid in the battle to flag and reduce sexist content, helping to lower its spread and impact.

As part of the NLP course assignment, we focused on addressing Task A - Binary Sexism Detection from the SemEval 2023 Task 10 challenge. This involves predicting whether a given text is sexist or not, leveraging the techniques and methodologies covered in the course to develop and evaluate our solution.

## 2 System description

For the needs of this assignment we use two pre-trained LLM models from the open-source repository for models of HuggingFace: the models Mistral v3 and Phi3-mini. Our task consisted of preparing prompts using given prompt templates and running inference on the selected models. Afterwards, an analysis and comparison of the results is done.

Mistral-7B-Instruct-v0.3 is an open-source language model developed by Mistral AI. It is an instruct fine-tuned version of the Mistral-7B-v0.3 model, which features an extended vocabulary of 32,768 tokens and supports a v3 tokenizer and function calling. (Mistralai, 2025)

Phi-3-mini-4k-instruct is a 3.8 billion parameter language model developed by Microsoft. It was

trained on the Phi-3 datasets, which include both synthetic data and filtered publicly available websites, focusing on high-quality and reasoning-dense content. The model supports a context length of up to 4,000 tokens. (Microsoft, 2025)

## 3 Data

Data used in this assignment is a small test set version of the original [Explainable Detection of Online Sexism \(EDOS\)](#) dataset (10, 2023). It consists of two datasets *a2\_test.csv* with 300 samples used for inference and *demonstrations.csv* with 1000 samples used as examples for few-shot-prompting. Each dataset contains samples with a string formatted text and a string label ('sexist' or 'not sexist'). The data were provided by the subject's teaching staff.

The datasets are balanced with respect to the labels column, ensuring fair representation of all classes. Both datasets also show a similar distribution of text lengths, with comparable minimum and maximum values and very close mean lengths.

Since we used pre-trained transformer models that handle much of the text cleaning process themselves, no additional preprocessing was necessary. The text was tokenized using the tokenizer provided by the pre-trained model.

## 4 Experimental setup and results

The goal of our experiments was to compare the performance between the models, between zero-shot and few-shot prompting, as well as evaluate the impact of changing the temperature settings in zero-shot prompting. We focused on understanding how these configurations influence both task performance and complexity.

Inference is calculated for both of the models using the following configurations:

1. zero-shot-prompting and temperature = None (default value = 1)

2. zero-shot-prompting and temperature = 0.3
3. few-shot-prompting (number of examples per class = 2)
4. few-shot-prompting (number of examples per class = 3)
5. few-shot-prompting (number of examples per class = 4)

The code for this assignment was written in Python, utilizing the models and datasets described in the previous sections. It was developed following the tutorials provided in the course materials. It is submitted alongside this report and is also available in the linked GitHub repository. The results are shown in Table 1.

## 5 Discussion

| Model name                   | Accuracy | Fail Ratio |
|------------------------------|----------|------------|
| Mistral v3, zero-shot, t=1   | 0.59     | 0.11       |
| Mistral v3, zero-shot, t=0.3 | 0.59     | 0.11       |
| Mistral v3, few-shot, n=2    | 0.77     | 0.36       |
| Mistral v3, few-shot, n=3    | 0.75     | 0.45       |
| Mistral v3, few-shot, n=4    | 0.71     | 0.28       |
| Phi3-mini, zero-shot, t=1    | 0.64     | 0.46       |
| Phi3-mini, zero-shot, t=0.3  | 0.64     | 0.46       |
| Phi3-mini, few-shot, n=2     | 0.62     | 0.70       |
| Phi3-mini, few-shot, n=3     | 0.64     | 0.59       |
| Phi3-mini, few-shot, n=4     | 0.66     | 0.55       |

Table 1: Table showing the performance of models under various configurations, where  $t$  represents the temperature setting and  $n$  denotes the number of examples per class used in few-shot prompting

As seen in our experiments, zero-shot prompting is faster due to its simplicity. Since it is an approach that doesn't rely on labeled examples, it is easy to implement and it makes an efficient baseline. In contrast, few-shot prompting took more time because it incorporates labeled examples into the prompt, providing more task-specific context. This added expressiveness and in the case of Mistral model, resulted in better performance, as shown in our results. However when changing the number of examples included in the prompt we can notice, after a certain point, the additional examples may not add much value and might even result in a slight decrease in accuracy. This could be due to the model becoming too constrained by the prompt and the result of the model overfitting.

When comparing the influence of the temperature parameter, in binary decision-making tasks, a low temperature (e.g., 0) is preferred for consistent and reliable outputs, as it reduces randomness. Higher temperatures are better for creative tasks, but a slightly higher temperature (e.g., 0.3) can capture subtler nuances, which is why we included it in our experiments. The default temperature of 1.0 balances randomness and confidence in predictions, while 0.3 sharpens the output, making the model more confident and deterministic. From the experiments ran we can conclude that the temperature setting may not significantly influence the accuracy or fail ratio for this task. Since the pre-trained models already generalize well, temperature changes have little impact, suggesting that the task might be straightforward or the model already produces high-confidence predictions.

When comparing the models between each other, we observe that Mistral v3 begins with a zero-shot approach that shows a bias toward the sexist class, but by adding examples in a few-shot setting, it converges to a more balanced and accurate solution. This significant improvement with few-shot prompting is likely due to Mistral's training on a dataset or architecture that benefits from contextual examples. In contrast, Phi3-mini shows little difference between few-shot and zero-shot performance, likely due to its architecture or pretraining being less responsive to contextual examples. Although Phi3-mini maintains a relatively consistent accuracy that is not far behind Mistral, it exhibits a higher fail ratio, indicating less reliability in predictions across scenarios.

## 6 Conclusion

Prompting approaches on pretrained models can provide decent results on straightforward tasks like this. Mistral v3 appears more fit for the task, especially in scenarios where few-shot prompting is feasible, as it delivers more reliable and accurate results. Phi3-mini, on the other hand, might be acceptable for less critical applications or when computational resources are limited, but its higher fail ratio requires more caution when it is used. Another approach worth exploring is fine-tuning these models and rerunning the experiment to see if performance improves further with task-specific training.

## 7 Links to external resources

- [GitHub repository](#)

## References

SemEval-2023 Task 10. 2023. [Comments: Semeval-2023 task 10 \(acl 2023\)](#). *arXiv preprint arXiv:2303.04222*. Accessed: 2025-01-12.

Microsoft. 2025. [Phi-3-mini-4k-instruct](#). Accessed: 2025-01-12.

Mistralai. 2025. [Mistral-7b-instruct-v0.3](#). Accessed: 2025-01-12.